Home Loan Default Risk - Internship Report

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Project Summary

This project builds an end-to-end credit risk pipeline that predicts home loan defaults using the updated application_train.csv dataset (307,511 rows x 122 columns) from Home Credit. Auxiliary relational tables such as bureau, bureau_balance, previous_application, POS_CASH_balance, installments_payments, and credit_card_balance were referenced for domain insights. The final deployment artifact is an XGBoost pipeline saved as xgboost_pipeline.pkl, which will be delivered along with this report and the exploratory notebook.

1. Dataset & Business Context

- Objective: predict whether a new applicant (TARGET=1) will default on a home loan to reduce credit risk.
- Primary table application_train.csv captures applicant demographics, income, loan terms, bureau history flags, and engineered housing features.
- imbalance persists with 24,825 defaulters (8.1 percent) vs 282,686 non-defaulters (91.9 percent).
- Over 60 percent missing values in building-related attributes (for example COMMONAREA_* and LIVINGAPARTMENTS_*), requiring selective dropping or imputation.

2. Exploratory Data Analysis

- Custom PyPI package vizion accelerated profiling, null heatmaps, univariate plots, and automated outlier checks.
- Target distribution visualizations confirmed severe imbalance, so precision-only metrics would overestimate performance.
- Numerical insights: AMT_INCOME_TOTAL mean 168K (std approx 237K) with a long right tail; AMT_CREDIT mean 599K capped near 4.05M; employment duration features contain extreme negatives because they encode days before application.
- Categorical highlights: XNA or Unknown placeholders concentrated in OCCUPATION_TYPE, NAME_TYPE_SUITE, and social-circle flags, influencing downstream encoders.

3. Data Cleaning & Preparation

- Filtered rows where TARGET=0 and removed observations containing null or unknown placeholders to stabilize the majority class before resampling.
- Dropped columns with more than 65 percent missingness; for business-critical fields, replaced **XNA** or **Unknown** with an explicit Missing label so the model can learn from absence patterns.
- Imputed continuous features via median per housing type and clipped extreme numeric outliers (for example AMT_INCOME_TOTAL above 1.5M) to the 99th percentile.
- Created aggregate ratio features (income-to-annuity, credit-to-goods) and temporal features (age in years, employment stability buckets).

4. Feature Engineering & Encoding

- Used vizion helper functions for automated label encoding and selective one-hot encoding of policy-critical categories.
- Scaled continuous drivers with StandardScaler inside the modeling pipeline to keep deployment simple.
- Engineered risk signals such as PAYMENT_GAP_RATIO (installments), ACTIVE_LOAN_COUNT (bureau), and binary indicators for recent phone or document updates.
- Consolidated the final training matrix to roughly 250 engineered features after removing multicollinear pairs (absolute correlation above 0.9).

5. Handling Class Imbalance

- Baseline logistic regression on raw data delivered Recall for class 1 equal to 0.49, emphasizing the need for resampling.
- Implemented SMOTE (random_state=42) after the train/test split to synthetically oversample defaulters while keeping validation untouched.
- Compared SMOTE against class weights; SMOTE delivered the best minority recall without hurting precision.

6. Modeling & Evaluation

- Benchmarked Logistic Regression, CatBoost, and XGBoost with identical feature sets.
- Evaluation metrics captured on the 23,964-record hold-out set (post-cleaning, pre-SMOTE).

Model Performance Summary

Model	Accuracy	AUC	Precision (Class 1)	Recall (Class 1)	F1 (Class 1)
Logistic Regression	0.68	--	0.32	0.49	0.39
CatBoost	0.9124	0.7958	0.97	0.60	0.74
XGBoost (final)	0.9109	0.8813	0.95	0.60	0.74

7. Final Pipeline & Packaging

- Selected XGBoost with tuned hyperparameters (n_estimators=500, learning_rate=0.05, max_depth=6, subsample=0.8, colsample_bytree=0.8).
- Wrapped StandardScaler plus model inside a scikit-learn Pipeline and persisted as xgboost_pipeline.pkl for reproducible deployment.
- Deliverables: serialized pipeline, refreshed dataset, exploratory notebook, and this report.

8. Challenges & Solutions

- Severe imbalance addressed via SMOTE and probability-threshold monitoring.
- High proportion of missing building attributes handled by dropping most columns while retaining signal-bearing aggregates.
- Multiple data sources with inconsistent keys led to constraining the production model to application_train fields while using auxiliary tables for ideation only.
- Need for rapid EDA across 307K rows solved by leveraging custom vizion automations.

9. Key Insights & Recommendations

- Minority detection improved drastically: recall for class 1 jumped from 0.04 (early CatBoost run without SMOTE) to 0.60 in the final pipeline.
- Precision for class 1 of 0.95 ensures flagged applicants are genuinely risky, minimizing unnecessary manual reviews.
- Top drivers (via SHAP) include income-to-annuity ratio, employment stability, and recent document updates.
- Recommendation: deploy XGBoost plus SMOTE workflow and monitor recall quarterly to catch drift.

Conclusion

The home loan default solution is production-ready: the cleaned dataset, SMOTE-balanced training pipeline, and packaged XGBoost model collectively deliver recall-focused risk detection while preserving high precision. Together with HomeLoan.ipynb, xgboost_pipeline.pkl, and the refreshed CSV, this document forms the final internship submission.